

Nutrition, Health Promotion and Counseling in the Prevention of Cardiovascular Disease



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Cardiovascular Disease: Then and Now



AMERICAN
COLLEGE of
CARDIOLOGY

**CARDIOVASCULAR
DISEASE**
AROUND THE
WORLD

CVD RISK FACTORS:

Obesity:
2008



1980



Hypertension:



Dyslipidemia:

An estimated 39% of the global population has elevated cholesterol



Smoking:
U.S. smoking rates have decreased, but at least 25% of residents living in Greece, Chile, Ireland, Hungary, Estonia, Spain, and Turkey smoke cigarettes



CURRENT LANDSCAPE:

Cardiovascular disease (CVD) is the leading global cause of death accounting for 17.3 million deaths/year

23.6 million

17.3 million

CVD deaths/year are expected to grow to more than 23.6 million by 2030

80% of CVD deaths occur in low- and middle-income countries

CVD accounts for nearly

one half of noncommunicable diseases (NCDs)

Source: The Worldwide Environment of Cardiovascular Disease: Prevalence, Diagnosis, Therapy, and Policy Issues: A Report From the American College of Cardiology

Diabetes or prediabetes affects over half of US adults.

(based on 2017 to 2020 data)



224.3 per 100,000

The age-adjusted US death rate attributable to CVD (based on 2022 data)

235.2 per 100,000

The age-adjusted global death rate attributable to CVD (based on updated 2021 data)

Heart Disease & Stroke Statistics 2025 Update

The American Heart Association (AHA) 2025 Statistics Update is a major source for monitoring cardiovascular health, risk factors, and disease in the US and global population. It is published annually in *Circulation* and can be accessed for free at AHAjournals.org/StatUpdate

25.5% of US adults have high LDL-C (≥ 130 mg/dL) (based on 2017-2020 data)



3.65 million deaths globally were attributed to high LDL-C (based on updated 2021 data)



On average, someone in the US dies of CVD every 34 seconds 2580 US deaths from CVD each day (based on 2022 data)



On average, someone in the US dies of a stroke every 3 minutes and 11 seconds 453 US deaths from stroke each day (based on 2022 data)



Stroke deaths in the United States increased by 28.7% between 2012 and 2022.

Cardiovascular Disease Around the World

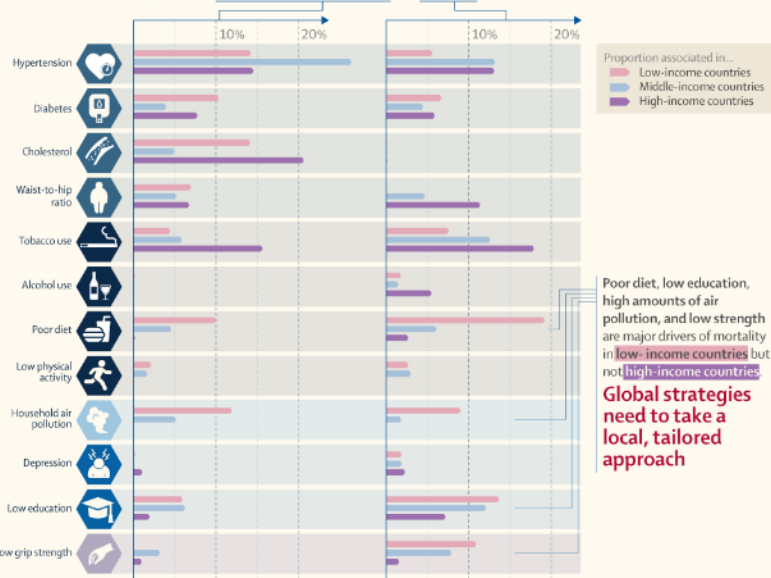


Over 70% of cardiovascular disease worldwide is caused by modifiable risk factors



But the causes are not the same around the world

Proportion of cardiovascular disease cases and mortality attributable to risk factors



GLOBAL HEARTS INITIATIVE
Working together
to beat cardiovascular disease

THE GLOBAL CARDIOVASCULAR DISEASE (CVD) CRISIS

Cardiovascular disease is the world's biggest killer, claiming **17.5 million** lives per year, 1/3 prematurely under 70 years.

This number is steadily increasing

80% of all CVD deaths are from heart attacks and strokes

75% occur in developing countries

Premature deaths from heart attacks and strokes threaten global health and development.

RISK FACTORS

Major risk factors contributing to CVDs are:



TOBACCO USE



CONSUMPTION OF FOODS
HIGH IN SALT



HIGH BLOOD PRESSURE

THE GLOBAL HEARTS INITIATIVE: Responding to the global cardiovascular disease crisis

An initiative to scale up national responses for prevention and management of cardiovascular diseases.



Technical package to defeat the global tobacco epidemic



Technical package for salt reduction



Technical package for cardiovascular disease management in primary health care

In the 2030 Sustainable Development Agenda, governments have committed to step up action to reduce premature deaths from cardiovascular disease and other NCDs to improve health and promote development.

www.who.int/global_hearts



For more, visit www.thelancet.com

- Modifiable risk factors, cardiovascular disease, and mortality in 155 722 individuals from 21 high-income, middle-income, and low-income countries (PURE)
- Variations in common diseases, hospital admissions, and deaths in middle-aged adults in 21 countries from five continents (PURE): a prospective cohort study

THE LANCET

The best science for better lives

Leading Causes of Cardiovascular Disease (CVD) Deaths

1. Ischemic Heart Disease
2. Stroke
3. Diabetes Mellitus
4. Hypertensive Heart Disease

Behavioral Risk Factors of CVDs

- Unhealthy diet
- Physical inactivity
- Tobacco use
- Harmful use of alcohol



Big Picture: 2026 AHA Dietary Guidance

- Focus on the overall dietary pattern across the life course
- Eat a wide variety of vegetables and fruits
- Choose mostly whole grains and healthy protein sources
- Replace saturated fats with unsaturated fats
- Choose minimally processed foods over ultra-processed foods
- Minimize added sugars and sodium
- If alcohol is not consumed, do not start; if consumed, limit intake

2026 Current Heart-Healthy Nutrition Targets

- **Dietary pattern**: emphasize vegetables, fruits, legumes, nuts, whole grains, fish/seafood, and liquid non-tropical plant oils
- **Saturated fat**: keep below 10% of energy; lower targets may be appropriate when LDL-C reduction is needed
- **Trans fat**: keep intake as low as possible
- **Sodium**: choose and prepare foods with little or no salt; aim for less than 2,300 mg/day
- **Added sugars**: minimize, especially sugar-sweetened beverages
- **Fiber**: prioritize naturally fiber-rich foods

FOODS THAT FIGHT INFLAMMATION

Chronic inflammation has been linked to cancer, heart disease, diabetes, arthritis, depression, and Alzheimer's. Fight inflammation with a healthy diet.



ANTI-INFLAMMATION FOODS



Tomatoes



Fruits

Strawberries, blueberries, oranges and cherries.



Nuts

Almonds, walnuts, and other nuts.



Olive oil



Leafy greens

Spinach, kale, collards, and more.



Fatty fish

Salmon, mackerel, tuna, and sardines.

INFLAMMATION FOODS



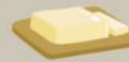
Fried foods



Sodas



Refined carbs



Lard



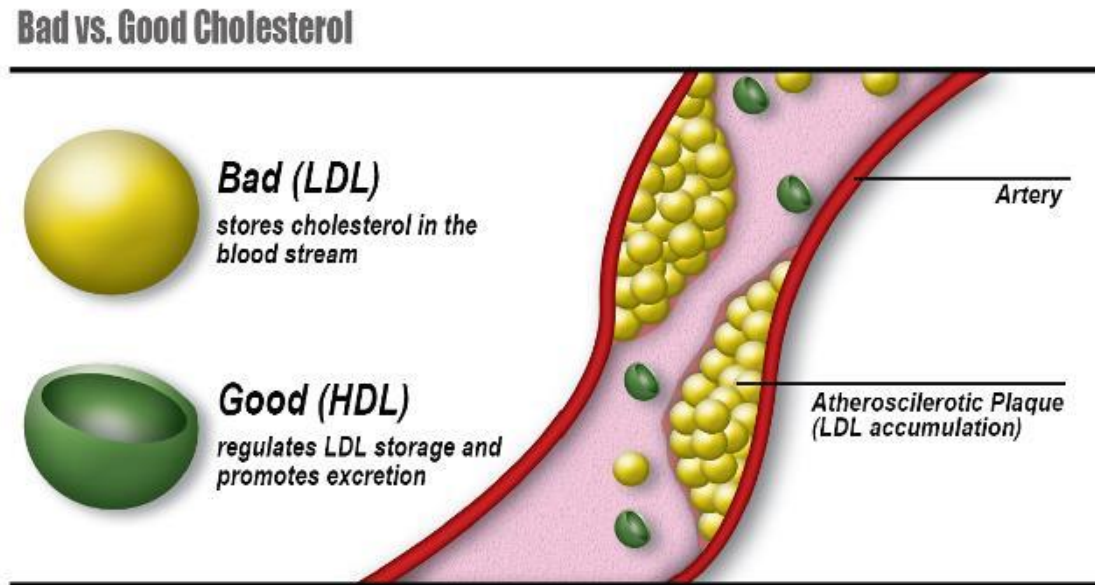
Processed meats

CVD		Prevention	Treatment
		INCREASE Vegetables Legumes Fish Fruits (<i>limited</i>) Olive oil (80 gram/day) Whole grain (< 100 gram/day) Nuts (30 gram/day) Time-restricted eating (11- 12 hours) Periodic Fasting-mimicking diet Exercise AVOID Red meat (<i>particularly processed</i>) Sugar (> 10 gram/day) Dairy Protein (> 0.36 gram/pound of bodyweight until age 65-70) Saturated and Trans fats Meal frequency (> 3 meals/day) Overweight/Obesity Stress	Longevity diet: ➤ Vegan/pescetarian-based diet ➤ No dairy ➤ Low fruit and fruit-juices ➤ Yes fish and extra-virgin olive oil ➤ Yes grains, legumes, lentils, vegetables ➤ Yes nuts (walnuts, hazelnuts, almonds) ➤ No caloric restriction (except overweight/ obese) ➤ Time-restricted eating (11-12 hours). ➤ 60% carbohydrate, 30% fat, 10% protein Ornish diet: ➤ Vegetarian diet ➤ 10% of calories as fat, 15-20% protein, 70-75% predominantly complex carbohydrates ➤ No caloric restriction ➤ Lasting > 1 year Esselstyn diet: ➤ Vegetarian diet ➤ No fish, oil, or dairy ➤ Consume grains, legumes, lentils, vegetables, and fruit ➤ No caloric restriction

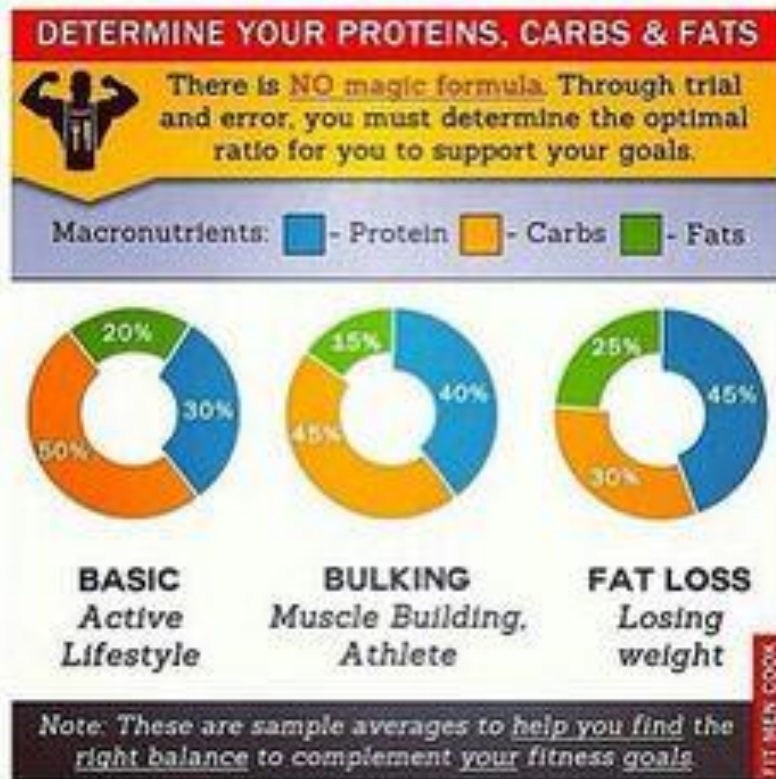


Cholesterol: LDL vs. HDL

- We want to keep LDL (“lousy”) low and HDL (“hero”) high.



Macronutrients



Carbohydrates

Breads
Rice
Couscous
Cereals
Bran
Potatoes
Pasta
Oats
Cream of Wheat
Corn
English Muffins
Pancakes
Whole Wheat/
Whole Grains
Vegetables
Squash
Pumpkin
Berries
Fruits
Sugars

Beans
Sprouted
Grains
Quinoa
Most Yogurts
Skim Milk
Peas

Macro Cheat Sheet

Proteins

Chicken
Turkey
Egg Whites
Fish
Buffalo
Bison
Whey Protein
Turkey Bacon
Lean Beef
Low/Non-fat
cottage
cheese
Low/Non-fat
greek yogurt

Eggs
Salmon
Bacon
Chia Seeds
Cottage
Cheese
Whole Fat
Milk
Duck
Whole-Fat
Yogurt

Acocado
Nut Butters
Egg Yolks
Nuts
Oils
Olives
Flaxseed

Fats

FATS

THE GOOD THE BAD & THE UGLY

✓ GOOD

Monounsaturated & Polyunsaturated Fats

- Can lower bad cholesterol levels
- Can lower risk of heart disease & stroke
- Can provide essential fats that your body needs but can't produce itself

SOURCE

Plant-based liquid oils, nuts, seeds and fatty fish

EXAMPLES



Oils (such as canola, olive, peanut, safflower and sesame)



Avocados



Fatty Fish (such as tuna, herring, lake trout, mackerel, salmon and sardines)



Nuts & Seeds (such as flaxseed, sunflower seeds and walnuts)

✗ BAD

Saturated Fats

- Can raise bad cholesterol levels
- Can lower good cholesterol levels
- Can increase risk of heart disease & stroke

SOURCE

Most saturated fats come from animal sources, including meat and dairy, and from tropical oils

EXAMPLES



Beef, Pork & Chicken Fat



Butter



Cheese (such as whole milk cheeses)



Tropical Oils (such as coconut, palm kernel and palm oils)

✗ UGLY

Hydrogenated Oils & *Trans* Fats

- Can raise bad cholesterol levels
- Can lower good cholesterol levels
- Can increase risk of heart disease & stroke
- Can increase risk of type 2 diabetes

SOURCE

Processed foods made with partially hydrogenated oils

EXAMPLES



Partially Hydrogenated Oils



Some Baked Goods



Fried Foods



Stick of Margarine

American Heart Association
Recommendation

Eat a healthy dietary pattern that:

Includes
good fats

Limits
saturated fats

Keeps trans fats as
LOW as possible

For more information, go to heart.org/fats

Saturated Fatty Acids (SFA)

- Butter, Lard
- Regular Milk
- Cheese
- Red Meat
- Breakfast Meats (sausage, bacon)
- Fast foods
- Processed foods (chips, cookies, etc)



Trans Fat: Current Clinical Message

- Industrially produced trans fats raise LDL-C and increase cardiovascular risk.
- Current practical message: keep trans-fat intake as low as possible.
- Check ingredient lists for “partially hydrogenated oil.”
- Small amounts of naturally occurring trans fat are present in meat and dairy from ruminant animals.
- Common sources may include certain fried foods, baked goods, shortenings, and imported or specialty products made with partially hydrogenated oils.

Saturated vs. Trans-Fat Effect on Lipid Panel

- Increases **total** and **LDL-C** cholesterol and decreases **HDL**
- Effect of Dietary trans Fatty Acids on High-Density and Low-Density Lipoprotein Cholesterol Levels in Healthy Subjects*

	Trans fat	Saturated fat
Total cholesterol	+6%	+12%
LDL cholesterol	+14%	+18%
HDL cholesterol	-12%	0%
LDL/HDL ratio	+29%	+18%

Are There Any “Healthy Fats”?



Omega-3 Polyunsaturated Fatty Acids (PUFAs)

EPA and DHA

- Prefer fish and seafood as part of a heart-healthy dietary pattern.
- Aim for approximately 2 servings of fish per week, especially fatty fish.

ALA

- Plant sources include walnuts, flaxseed, chia seeds, and canola or soybean oil.

Supplement counseling

- Routine over-the-counter fish-oil supplementation is not recommended for primary prevention of CVD.
- High-dose prescription EPA may be considered for selected high-risk patients with persistent hypertriglyceridemia under clinician supervision.
- Discuss bleeding risk, atrial fibrillation risk, product quality, and drug interactions.

Omega-3 Polyunsaturated Fatty Acids (PUFAs)



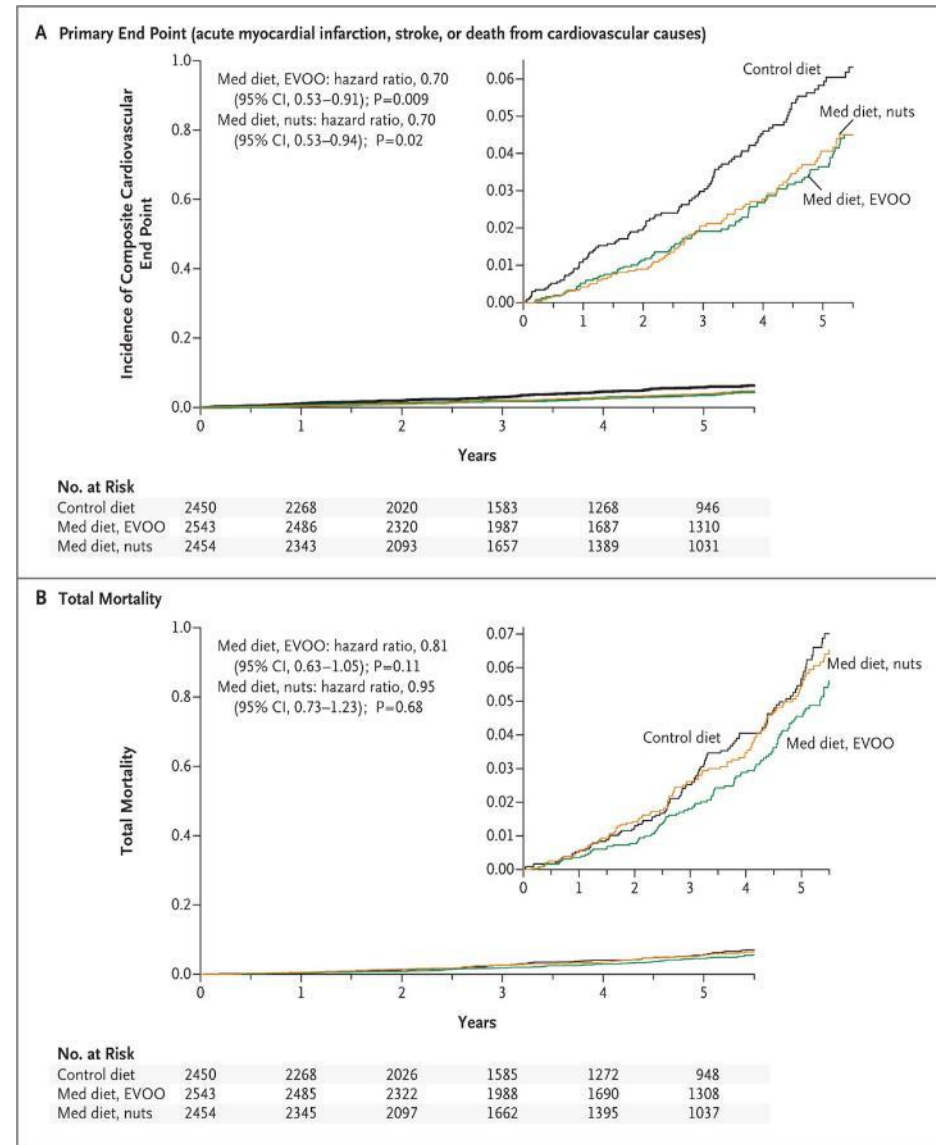
Monounsaturated Fatty Acids (MUFAs)

- **Basis of Mediterranean Diet:**
 - High in MUFA and low in SFA
→ decreased risk of CVD.
 - Emphasizes fruits, root vegetables, flax seeds, and **MUFAs**.
 - *Avocado, olives, nuts, olive oil and canola oil.*



Mediterranean Diet

- Primary Prevention of Cardiovascular Disease with a Mediterranean Diet (PREDIMED study)
- Among persons at **high cardiovascular risk**, a Mediterranean diet supplemented with extra-virgin olive oil or nuts **reduced the incidence** of major cardiovascular events.



- Ingestion of soluble fiber in legumes, oats, fruit and vegetables → 10-14% reduction in LDL-C.



Nutritional Guidance for Managing Hypertension

- **Consume:**

- Vegetables, fruits, whole grains
- Low-fat dairy products
- Poultry, fish, legumes, vegetable oils, nuts

- **Limit:**

- Sweets and sugar-sweetened beverages
- Red meats, saturated and trans-fats
- **Sodium (salt)**

Lower Sodium Intake

- Most dietary sodium comes from packaged, prepared, and restaurant foods, not the salt shaker.
- Clinical target: less than 2,300 mg sodium/day for most adults.
- Greater reductions may further lower blood pressure for many patients when nutritionally appropriate and feasible.

Practical strategies:

- • Compare Nutrition Facts labels
- • Choose “no salt added” or lower-sodium options
- • Flavor foods with herbs, spices, citrus, or vinegar
- • Use DASH-style eating patterns

Combine DASH Diet With Lower Sodium Intake

- **DASH** = Dietary Approach to Stop Hypertension
 - **High** in vegetables, fruits, low-fat dairy products, whole grains, poultry, fish, and nuts.
 - **Rich** in potassium, magnesium, calcium, lean protein, and soluble fiber.
 - **Low** in sweets, sugar-sweetened beverages, and red meats.
 - **Low** in saturated fat, total fat, and cholesterol.

Funny, They Don't Taste Salty



TONY CENICOLA/THE NEW YORK TIMES

	COOKIES	WAFFLES	CHICKEN BREAST	GRANOLA BARS	SYRUP	WHEAT BREAD	PITA BREAD	SALAD DRESSING	COTTAGE CHEESE
SERVING	Three Oreos.	Two Eggo waffles.	1/2 cup Perdue, roasted.	Two Nature Valley Oats 'n Honey bars.	1/4 cup Eggo Lite syrup.	1 slice Arnold whole wheat.	1 Damascus Bakeries whole wheat pita.	Two table-spoons Kraft fat-free Italian.	1/2 cup Light n' Lively low-fat.
SALT	160 mg.	410 mg.	460 mg.	160 mg.	180 mg.	170 mg.	290 mg.	480 mg.	420 mg.
PERCENT OF DAILY INTAKE*	11%	27%	31%	11%	12%	11%	19%	32%	28%

*Percentages are calculated from the Institute of Medicine report that recommends 1,500 milligrams of salt each day as adequate for people ages 9-50; an older government standard of a 2,400 milligram daily intake appears on product labels, with correspondingly lower percentages.

Insulin Resistance → CVD Risk Factor

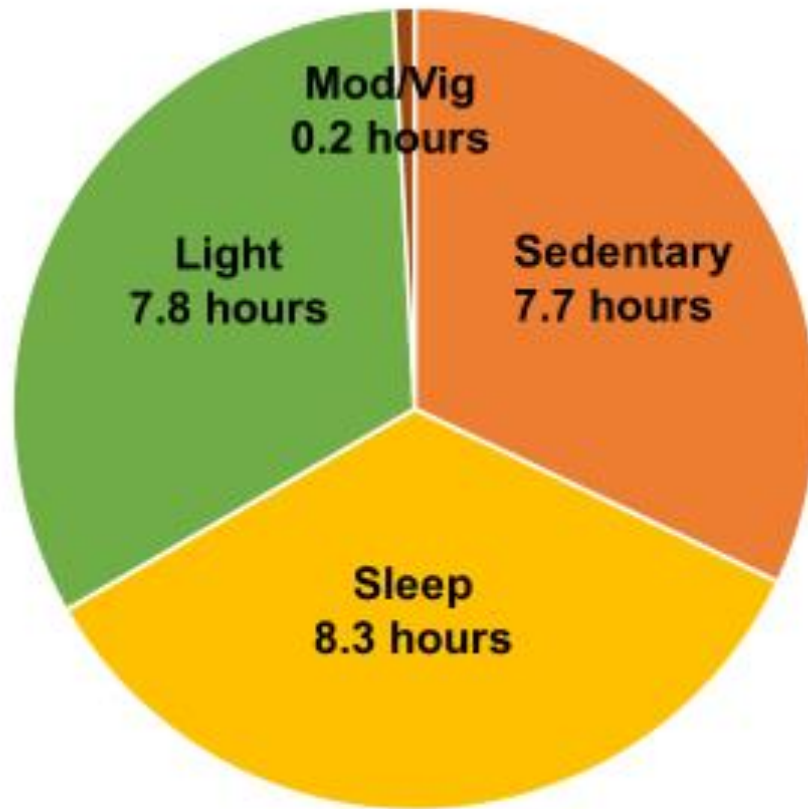
- Metabolic Syndrome diagnosis → 3 of 5 criteria:

Elevated blood glucose	≥100 mg/dL or receiving drug therapy for hyperglycemia
Abdominal obesity	Waist circumference ≥40" in men or ≥ 35" in women
Elevated BP	≥130/85 mm Hg or receiving drug therapy for BP
Elevated TG	≥150 mg/dL or receiving drug therapy for elevated TG
Reduced HDL-C	< 40 mg/dL in men or < 50 mg/dL in women, or receiving drug therapy for reduced HDL-C

Steps to Promote Optimal Weight and Nutrition

- Goal: Reduce ASCVD risk
 - Measure BMI and waist circumference.
 - Discuss findings with the patient.
 - Assess patient's motivation to change.
 - Assess dietary intake.
 - 24-hour recall of their diet
 - Assess other lifestyle choices:
 - Tobacco use
 - Alcohol use
 - Physical activity level
 - Counsel patient about nutrition and set small goals to help them reach the main goal

Hours Per Day Spent in Various States of Activity



U.S. adults spend >7 h/d on average in sedentary activities. Replacing sedentary time with other physical activity involves increasing either moderate- to vigorous-intensity physical activity or light-intensity physical activity.

NEW LABEL / WHAT'S DIFFERENT

Servings:
larger,
bolder type

Nutrition Facts

8 servings per container

Serving size 2/3 cup (55g)

Amount per serving

Calories 230

% Daily Value*

Total Fat 8g 10%

Saturated Fat 1g 5%

Trans Fat 0g

Cholesterol 0mg 0%

Sodium 160mg 7%

Total Carbohydrate 37g 13%

Dietary Fiber 4g 14%

Total Sugars 12g

Includes 10g Added Sugars 20%

Protein 3g

Vitamin D 2mcg 10%

Calcium 260mg 20%

Iron 8mg 45%

Potassium 235mg 6%

* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

— Serving sizes
updated

— Calories:
larger type

— Updated
daily
values

— Actual
amounts
declared

— New
footnote

— New:
added sugars

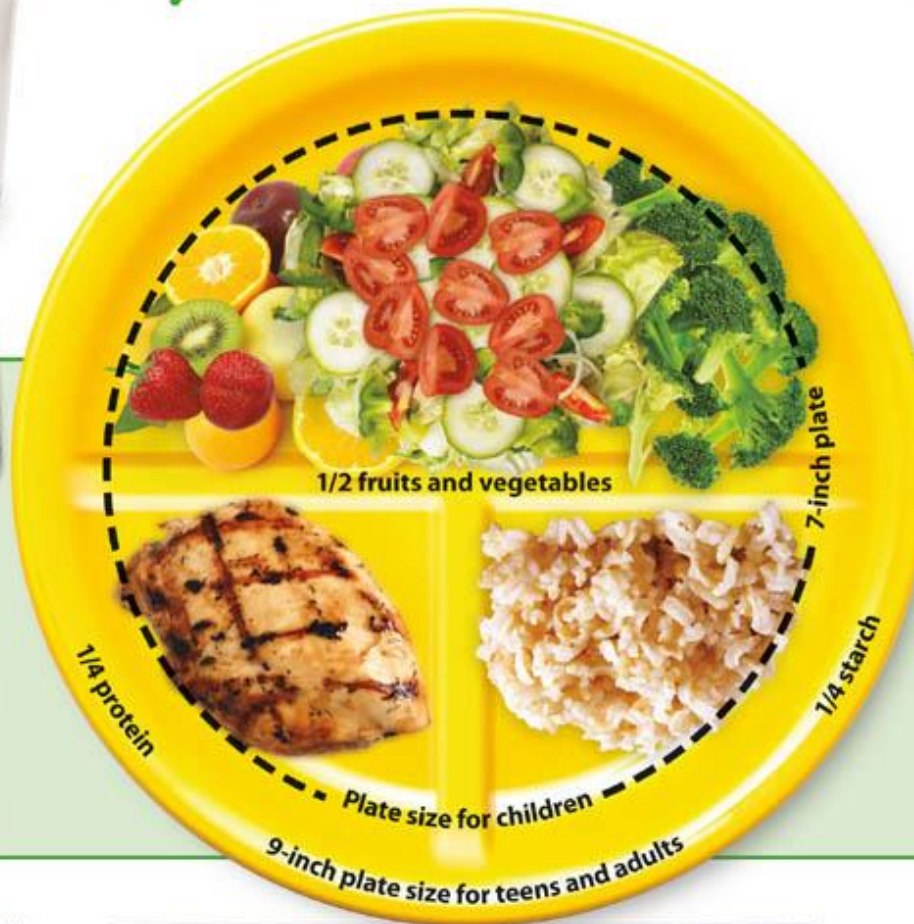
— Change
in nutrients
required

My Plate Planner



Give children a small plate (or small portions on a big plate).

Split the plate into 3 parts, the largest for fruits and vegetables.



Use your hand to measure the right amount of food to eat. Use an adult hand for adult portions. Use a child's hand for children's portions.

 Palm of Hand
Amount of lean meat

 A Fist
Amount of rice, cooked pasta, or cereal

 A Thumb
Amount of cheese

 Thumb Tip
Amount of salad dressing



your health matters

Adapted with permission from the New York City Department of Health and Mental Hygiene.
HHP/CHS/10/04 | 11.2010



◀ 7-inch plate for children

◀ 9-inch plate for teens and adults

Key Takeaways

- Counsel to adopt a sustainable heart-healthy dietary pattern across the life course.
- Emphasize:
 - Vegetables and fruits, whole grains, legumes, nuts, healthy protein sources, and unsaturated fats
 - Minimally processed foods and culturally appropriate choices
- Minimize:
 - Ultra-processed foods, added sugars, excess sodium, trans fat, and foods high in saturated fat
- Prefer fish/seafood over routine fish-oil supplements for primary prevention.
- Use shared decision-making, small achievable goals, and follow-up to support behavior change.

Lecture Feedback Form

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

Thank You 😊

